

Supplementary Material for “Regime-Adaptive Risk-Aware MPPI Control for Force-Aware Robotic Wiping with Temporal-Difference World Models”

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I. METHOD DEVELOPMENT RESULTS

A. Adaptive Objective and Compute

All four planner arms completed all 60 development evaluations, and no native sample exceeded 15 N. Table S1 summarizes the matched results across the three force targets. M4 has the lowest mean MRE and NRMSE at every target. Objective adaptation and compute adaptation each recover part of the difference from M1, while their combination gives the largest 12 N improvement. M4 reduces 12 N RMSE from 2.927 to 2.600 N and raises mean force from 9.308 to 9.714 N. The strict 12 N tracking gate nevertheless remains 0/5 for every arm.

The compute schedule lowers the mean sample count because stable acquisition and lower-force tracking often use a smaller budget. At 12 N, the schedule increases the sample count when predicted transient risk rises. The complete method therefore allocates computation by regime rather than reducing it uniformly.

B. Behaviour-Cloning Strength

The one-step force-prediction RMSE is effectively unchanged across BC coefficients, whereas actor-teacher action RMSE falls from 1.006 without BC to 0.099 and 0.090 at coefficients 0.5 and 2. Table S2 shows the resulting closed-loop difference. Without BC, none of the 15 runs establishes contact. A coefficient of 0.5 restores contact but remains seed-sensitive, with three task failures and two samples above 15 N. The coefficient of 2 completes all runs without a sampled violation. BC therefore stabilizes the actor proposal used by MPPI rather than improving the one-step force model.

C. Router and Risk-Representation Ablations

The historical hard router was evaluated at six activation fractions from 0.70 to 0.95 across the same five checkpoints and three force targets. All 90 runs completed without a sampled force violation. The actor-step fraction rose from 27.1% to 60.5%, whereas mean MRE changed gradually from 6.45% to 5.25%. At 12 N, actor use rose from 25.2% to 88.1%, yet mean MRE remained 10.16% at the highest threshold. There is therefore no discrete operating point that resolves the high-force limitation; the threshold principally controls planner use.

The earlier registered-frame controller also permits a sample-pooled decomposition of its 56,481 contact samples at

12 N. Stable MPPI operation accounts for 51,771 samples with a mean force of 11.020 N, actor re-entry accounts for 1,950 samples at 10.075 N, and initial or other contact accounts for 2,760 samples at 6.605 N. Their pooled mean is 10.772 N; the corresponding mean of per-evaluation contact means is 10.767 N. The stable planner segment therefore contains most of the negative bias, while actor re-entry deepens it. This evidence motivated separate router and risk-representation ablations rather than attributing the 12 N shortfall to contact loss alone.

The three risk representations each complete 15/15 runs, pass strict tracking in 10/15, and produce no sample above 15 N. Force-only and point-envelope trajectories are identical. The calibrated alternative changes only the 12 N trajectories and gives a slightly higher aggregate NRMSE (12.94% versus 12.91%). The frozen rule therefore selects next-force prediction alone. These development results select the full M4 objective-and-compute mechanism, BC coefficient 2, and the force-only risk representation before the factor-separated evaluation.

II. PROVENANCE AND NUMERICAL AUDIT

The preceding experiments answer whether BC-assisted direct TD-MPC2 can complete the evaluated task and where its behaviour changes. This section asks a different question: which parts of that evidence retain the same interpretation after the geometric sources and numerical settings are inspected?

For the 288 matched-OOD evaluations, a separate audit program re-read the frozen JSONL traces, verified their hashes, and recomputed all 7,776 reported per-evaluation metric values from an independently implemented specification. The program imports neither the experimental analysis nor the finalizer or runner. Every value matched the stored summary, with no trace-hash or trace-identity mismatch. Here, independent refers to the metric-computation logic, not to an independently generated trajectory set.

A. Geometric and Disturbance Provenance

Table S3 separates information available before action selection from information that would be available through a physical sensing stack. The controller is time-causal, but it uses simulator state and a registered nominal surface model. The latter supplies the local frame for both observations and action execution; it is not inferred by TD-MPC2.

TABLE S1

MATCHED PLANNER-COMPONENT DEVELOPMENT ABLATION. MRE AND NRMSE ARE AVERAGED OVER 15 SEED-TARGET EVALUATIONS PER ARM.

Arm	MRE (%)	NRMSE (%)	Samples	Median time (ms)
M1: fixed objective, fixed compute	12.03	14.66	128.0	112.18
M2: adaptive objective	10.94	13.77	128.0	107.86
M3: adaptive compute	10.26	13.52	121.2	80.58
M4: complete method	9.65	12.94	120.4	83.25

TABLE S2

BEHAVIOUR-CLONING DEVELOPMENT ABLATION OVER FIVE SEEDS AND THREE FORCE TARGETS.

BC	Task	Tracking	No contact	Samples > 15	Max peak (N)
0	0/15	0/15	15/15	0	0.000
0.5	12/15	10/15	0/15	2	16.513
2	15/15	10/15	0/15	0	13.924

TABLE S3

SOURCES OF QUANTITIES USED BY THE CONTROLLER AND EVALUATOR.

Quantity	Source	Role and interpretation
Normal force and rate	Simulator contact readback	Observation, routing, and force metrics
Tool pose and velocity	Simulator state	Surface-relative observation
Path tangent and surface normal	Registered nominal model	Observation frame and Cartesian action map
Progress and contact dose	Synthetic task state	Observation and completion proxy

A separate claim-binding audit covered all controlled and OOD block labels. The physical call chains for the designated lateral and normal impulses were present. By contrast, the audit did not establish that the declared reference noise and delayed sensor noise in two controlled and two OOD blocks reached the learner-visible force path. Those labels are therefore not used as evidence of noise robustness.

B. Outcome-Specific Numerical Sensitivity

All four 12 N OOD baseline sentinels reproduced their archived traces within the prespecified tolerance. Supplementary Tables S2 and S3 expose all 32 paired intervention cells. Each force entry is the candidate peak in newtons, followed by three pass bits for task, tracking, and the force limit. The sentinel headers give the corresponding reproduced-baseline values. An arrow marks a trigger of the prespecified material peak/threshold rule and its direction; an unmarked value did not trigger the rule. Status changes relative to baseline are summarized below from the same paired cells.

The rule triggered in 11/32 cells. Task-success status changed in three cells, tracking status in two, and sampled force-limit status in two. Ten of the eleven material peak responses were reductions. The remaining response was an increase that crossed the 15 N threshold.

Aggregate pass-fail counts alone mask these cell-level outcome switches. Repeating each block’s baseline outcome across the eight interventions gives 16 task, 16 tracking, and 16 force-limit passes. The candidates give 15, 16, and 16, respectively, despite the paired cell-level changes in the complete cell tables in the Supplementary Material. The relevant finding is therefore outcome-specific sensitivity, not a binary division between invariant task results and unstable force results.

All twelve requested source arc counts reached the mesh factory, but PhysX convex cooking returned baseline-equivalent collision payloads in eleven cells (Supplementary Material). Only the 96-segment F4 request changed the topology, from 14 vertices and 24 triangles to 16 and 28, and produced the arm’s sole material peak response. The arm thus records request-to-collision mapping rather than multilevel mesh convergence. The 0.080 mm analytical source-mesh sagitta bound makes penetration-depth error unlikely as the sole explanation. Position iterations and contact offset also affect selected sentinels; physics substeps remain fixed in this 100 Hz, one-step panel.

C. Resulting Evidence Boundaries

Table S4 states the paper’s principal conclusions at the level supported by their evidence. The controlled experiments remain the main demonstration that the complete BC-assisted TD-MPC2 recipe can complete the evaluated ForceWipe task through direct deployment. The audit changes the interpretation of selected force-tail and disturbance claims; it does not replace that capability result.

APPENDIX A ARTIFACT PACKAGE

The public reproducibility package contains task and factor-block definitions, training and evaluation scripts, compact frozen results, bootstrap analyses, cross-simulator calibration records, numerical-sensitivity cells, and verification tests. Five checkpoints and the step-level evaluation traces are archived at <https://doi.org/10.5281/zenodo.22929983>. Repository and archive manifests record the relative path, byte count, and

TABLE S4

CLAIM-SPECIFIC EVIDENCE AFTER THE PROVENANCE AND NUMERICAL AUDIT. COUNTS REFER TO THE 225 CONTROLLED EVALUATIONS, 90 FULL-TD-MPC2 OOD EVALUATIONS, 45 CROSS-ENGINE EVALUATIONS, OR 32 PAIRED NUMERICAL-INTERVENTION CELLS IDENTIFIED IN EACH ROW.

Conclusion	Supported statement	Boundary
Controlled completion	BC-assisted TD-MPC2 completed 225/225 first-pass evaluations	Exact for the frozen controlled configuration; not a physical deployment claim
Controlled tracking	223/225 met the force-tracking criteria; the 12 N tier was biased low	Completion and precise regulation remain distinct
Weak-curvature transfer	Task and tracking passes fell to 39/90 and 37/90	Combined block shift; individual causes are not isolated
Force tails	Recorded peaks and exceedances describe the baseline numerical configuration	Material peak/threshold responses occurred in 11/32 12 N OOD sentinel cells
MPPI contribution	Task and tracking intervals crossed zero; host planning time increased by 50.96 ms	Same-checkpoint comparison under the frozen configuration; no general peak-force claim
Noise robustness	No claim retained for the four unverified reference/sensor-noise labels	Declared effects were not shown to reach the learner-visible path
Cross-engine portability	MuJoCo achieved 30/30 compound passes on flat and inclined surfaces but 0/15 on the cylinder	Zero-shot stress test under a documented tangential calibration mismatch; not solver equivalence or hardware evidence

TABLE S5

DIRECT PPO DEVELOPMENT SENSITIVITY OVER 90 EVALUATIONS PER ENDPOINT.

Endpoint	Compound	Task	Tracking	Safety
Default, 1×	0	8	4	89
Default, 2×	0	9	0	79
Default, 4×	1	4	8	87
LR 10^{-4} , 2×	1	16	6	87
LR 10^{-3} , 2×	1	4	7	70
Clip 0.1, 2×	3	11	6	79
Clip 0.3, 2×	0	13	5	88
Entropy 0, 2×	0	8	0	90
Entropy 0.01, 2×	0	5	7	89

SHA-256 digest of every payload file. Reproduction starts from the environment record and checkpoint manifest, then runs the frozen evaluators and independent analyses against the recorded trace keys.

APPENDIX B

DIRECT PPO CONFIGURATION

The direct PPO reference uses a 16-dimensional observation, a three-dimensional Cartesian action, two hidden layers of width 256, a learning rate of 3×10^{-4} , discount 0.99, generalized-advantage parameter 0.95, clip ratio 0.2, entropy coefficient 0.005, and value coefficient 0.5. Each update uses 2,048 rollout transitions, ten epochs, and eight minibatches per epoch, with gradient norm clipped at 0.5.

The budget study checkpoints the default setting at 82,774, 165,548, and 331,096 transitions. Six one-factor variants use 165,548 transitions and change learning rate to 10^{-4} or 10^{-3} , clip ratio to 0.1 or 0.3, or entropy coefficient to 0 or 0.01. Table S5 reports all development endpoints. Every endpoint contains 90 evaluations; authority passes are 90/90 throughout. The fixed selection rule chose clip ratio 0.1 at 2× budget. No final-suite result was used for selection.

APPENDIX C

CELL-LEVEL NUMERICAL-SENSITIVITY RESULTS

Tables S6 and S7 report every factor–sentinel cell used in the numerical-sensitivity analysis. The three-bit suffix records task, tracking, and sampled-force-limit pass status in that order.

The mesh factory received all 12 requested source resolutions. Ten cooked payloads matched the 48-segment baseline digest; the eight-segment F4 request retained the same 24-triangle connectivity, while its 14 vertex coordinates differed by at most 1.39×10^{-17} m. Only the 96-segment F4 request changed the topology and triggered the source-resolution peak rule.

TABLE S6

F1/F2 NUMERICAL-SENSITIVITY CELLS ($n = 8$ PAIRED INTERVENTIONS PER SENTINEL); ENTRIES GIVE PEAK FORCE (N) AND TASK/TRACKING/FORCE-LIMIT PASS CODES.

Intervention	F1: 24.038/000	F2: 14.849/111
Position iterations: 15 $\rightarrow$ 30	14.043↓/111	14.051/111
Velocity iterations: 1 $\rightarrow$ 4	22.971/000	13.766/111
Contact offset: 20 $\rightarrow$ 5 mm	16.589↓/000	15.421↑/000
Contact offset: 20 $\rightarrow$ 1 mm	17.852↓/000	14.028/111
Source-segment request: 48 $\rightarrow$ 24	24.038/000	14.849/111
Source-segment request: 48 $\rightarrow$ 96	24.038/000	14.849/111
Source-segment request: 48 $\rightarrow$ 8	24.038/000	14.849/111
1 mm offset and 8 segments	17.852↓/000	14.028/111
Material responses	4/8	1/8

TABLE S7

F4/F5 NUMERICAL-SENSITIVITY CELLS ($n = 8$ PAIRED INTERVENTIONS PER SENTINEL); ENTRIES GIVE PEAK FORCE (N) AND TASK/TRACKING/FORCE-LIMIT PASS CODES.

Intervention	F4: 24.926/000	F5: 13.630/111
Position iterations: 15 $\rightarrow$ 30	15.075↓/000	12.001↓/011
Velocity iterations: 1 $\rightarrow$ 4	23.282/000	12.814/111
Contact offset: 20 $\rightarrow$ 5 mm	15.673↓/000	13.066/111
Contact offset: 20 $\rightarrow$ 1 mm	18.806↓/000	13.407/111
Source-segment request: 48 $\rightarrow$ 24	24.926/000	13.630/111
Source-segment request: 48 $\rightarrow$ 96	21.579↓/000	13.630/111
Source-segment request: 48 $\rightarrow$ 8	24.926/000	13.630/111
1 mm offset and 8 segments	18.806↓/000	13.407/111
Material responses	5/8	1/8

APPENDIX D

CROSS-SIMULATOR CALIBRATION GATE

The MuJoCo port was calibrated without loading a policy checkpoint. Three contact-free motions identified a two-sample delay and 80-ms first-order actuator response. Four exposed contact motions then selected the contact and drive parameters. Table S8 gives the errors on two new held-out motions. The multilevel normal cycle meets every tolerance. The bidirectional tangential motion meets the force, onset, and peak tolerances, but its 0.833-mm position RMSE exceeds the 0.5-mm limit. The later policy run is therefore a zero-shot stress test rather than matched-engine replication.

TABLE S8

HELD-OUT CROSS-SIMULATOR TASK-EQUIVALENCE CALIBRATION.

Motion	F-NRMSE (%)	Pos. RMSE (mm)	Onset (samples)	Peak error (%)
Multilevel normal cycle	7.16	0.337	4	4.76
Bidirectional tangent	9.37	0.833	4	1.99
Tolerance	20.0	0.500	5	20.0